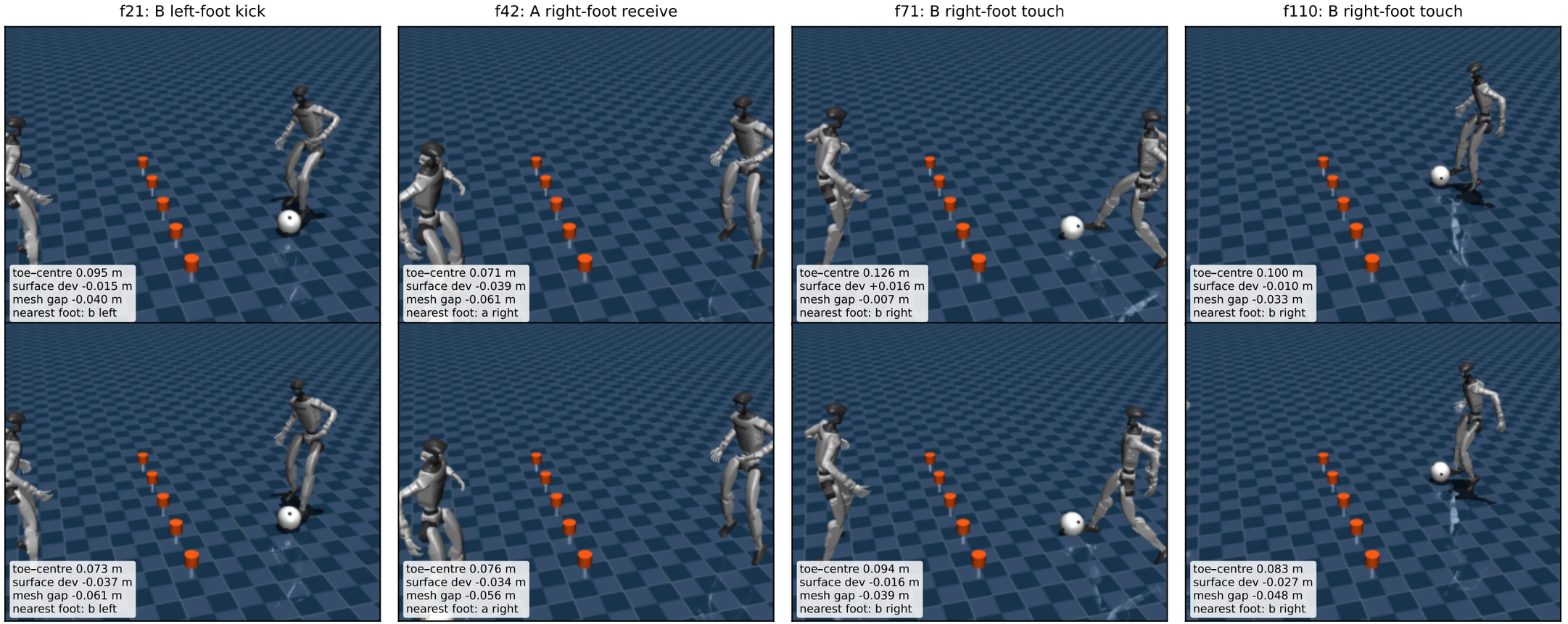
Fixed frozen ball + shared reference: event-conditioned IK OFF vs ON




Selected interaction frames for qualitative comparison; per-frame annotations are not aggregate scores. Same 144-frame ball path, same reference targets, same initialisation and iteration budget. Negative surface deviation / mesh gap = toe or foot mesh inside the 0.11 m ball (kinematic proxy, no physics).